

Master “Between Two Arrays” - Ek Complete Guide

Hinglish mein seekho, Real-Life Examples ke saath! 🚀

Page 1: Problem Ko Samjho (Understanding the Problem)

Problem Statement (Simple Words Mein)

Tumhare paas do arrays hain:

- **Array A:** Kuch numbers ka collection
- **Array B:** Kuch numbers ka collection

Tumhe aise numbers dhundhne hain jo:

1. Array A ke **har element ke multiples** hon (yaani Array A ke numbers unke factors hon)
2. Array B ke **har element ke factors** hon (yaani wo numbers Array B ke numbers ko perfectly divide karein)

Real-Life Analogy: 🏠

Socho tum school mein ho:

- Array A = [2, 4] → Ye chhote bachche hain (Junior students)
- Array B = [16, 32, 96] → Ye bade bachche hain (Senior students)
- Tumhe aise students dhundhne hain jo:
 - Juniors ke “seniors” hon (unse bade/experienced)
 - Seniors ke “juniors” hon (unse chhote/less experienced)

Keywords Samjho

Term	Simple Meaning	Real-Life Example
Factor	Jo number kisi doosre number ko perfectly divide kare	3, 12 ka factor hai kyunki $12 \div 3 = 4$ (no remainder)
Multiple	Kisi number ki table mein aane wala number	12, 3 ka multiple hai kyunki $3 \times 4 = 12$

Easy Trick: 🏠

- Factor = “Chhota bhai” (divides the bigger number)

- Multiple = “Bada bhai” (gets divided by smaller number)

Example: 4 aur 12 ke relationship mein:
4 → 12 ka factor hai (4 chhota hai, divide kar sakta hai)
12 → 4 ka multiple hai (12 bada hai, 4 ki table mein aata hai)

Page 2: Mathematical Foundation - Part 1

Factor (Gunankhand)

Definition: Factor wo number hai jo kisi doosre number ko completely divide kare, bilkul zero remainder ke saath.

Real-Life Analogy:  Pizza Slices

- Tumhare paas 12 pizza slices hain
- Tum unhe equally divide kar sakte ho:
 - 1 group mein 12 slices ($1 \times 12 = 12$) → 1 is a factor
 - 2 groups mein 6 slices ($2 \times 6 = 12$) → 2 is a factor
 - 3 groups mein 4 slices ($3 \times 4 = 12$) → 3 is a factor
 - 4 groups mein 3 slices → 4 is a factor
 - 6 groups mein 2 slices → 6 is a factor
 - 12 groups mein 1 slice → 12 is a factor

Factors of 12: [1, 2, 3, 4, 6, 12]

Multiple (Gunaj)

Definition: Multiple wo number hai jo kisi number ki table mein aata hai (number × kuch integer).

Real-Life Analogy:  Bus Schedule

- Bus har 3 minute mein aati hai
- Bus aayegi: 3, 6, 9, 12, 15, 18... minutes pe
- Ye sab 3 ke multiples hain!

Multiples of 3: [3, 6, 9, 12, 15, 18, 21, ...]

Factor vs Multiple - Ek Saath Samjho

Number = 6 ke liye:
Factors of 6: [1, 2, 3, 6] # Jo 6 ko divide karein

```
Multiples of 6: [6, 12, 18, 24, ...] # 6 jinko divide kare

# Relationship:
# 2 is a factor of 6 → 6 is a multiple of 2
# 3 is a factor of 6 → 6 is a multiple of 3
```

Page 3: Mathematical Foundation - Part 2

Greatest Common Divisor (GCD) - Mahatam Samapvartak

Definition: Do ya zyada numbers ka GCD wo **sabse bada number** hai jo un sabhi ko perfectly divide kar sake.

Real-Life Analogy:  Room Division

- Tumhare paas 12 boys aur 18 girls hain
- Tumhe groups banane hain jisme:
 - Har group mein same number of boys
 - Har group mein same number of girls
 - Maximum kitne groups ban sakte hain?

Boys: $12 = 2 \times 2 \times 3$
Girls: $18 = 2 \times 3 \times 3$
Common factors: 2, 3
 $GCD = 2 \times 3 = 6$ groups maximum!

Calculate GCD of [16, 32, 96]:

Step 1: GCD(16, 32)
 $16 = 2 \times 2 \times 2 \times 2$
 $32 = 2 \times 2 \times 2 \times 2 \times 2$
Common = $2 \times 2 \times 2 \times 2 = 16$ ✓

Step 2: GCD(16, 96)
 $16 = 2 \times 2 \times 2 \times 2$
 $96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$
Common = $2 \times 2 \times 2 \times 2 = 16$ ✓

Final GCD of [16, 32, 96] = 16

Least Common Multiple (LCM) - Laghutam Samapvartak

Definition: Do ya zyada numbers ka LCM wo **sabse chhota number** hai jo un sabhi ka multiple ho.

Real-Life Analogy:  Alarm Synchronization

- Alarm A har 4 minute mein bajta hai
- Alarm B har 6 minute mein bajta hai
- Dono ek saath kab bajenge?

Alarm A: 4, 8, 12, 16, 20, 24...

Alarm B: 6, 12, 18, 24...

First common: 12 minutes! (LCM)

Calculate LCM of [2, 4]:

Step 1: LCM(2, 4)
Multiples of 2: 2, 4, 6, 8...
Multiples of 4: 4, 8, 12...
LCM = 4 ✓

Shortcut formula: $LCM(a,b) = (a \times b) \div GCD(a,b)$
 $LCM(2,4) = (2 \times 4) \div 2 = 8 \div 2 = 4 \checkmark$

Page 4: Connection Between Concepts - Deep Dive

Why LCM for Array A? (First Array)

Logic: Jo number hum dhundh rahe hain, wo Array A ke har element ka multiple hona chahiye.

Array A = [2, 4]
Matlab answer aisa number hona chahiye jo:
- 2 ka multiple ho → 2 se divisible
- 4 ka multiple ho → 4 se divisible

Agar koi number 4 se divisible hai, toh wo automatically 2 se bhi divisible hai
But safest approach: LCM le lo jo GUARANTEE kare ki har element se divisible hai

$LCM(2,4) = 4$
Iska matlab: Answer 4 se divisible hona chahiye

Why GCD for Array B? (Second Array)

Logic: Jo number hum dhundh rahe hain, wo Array B ke har element ka factor hona chahiye.

Array B = [16, 32, 96]
Matlab answer aisa number hona chahiye jo:
- 16 ko divide kare → factor of 16
- 32 ko divide kare → factor of 32
- 96 ko divide kare → factor of 96

Agar koi number 16 ka factor hai, toh possible hai ki 32/96 ka factor na ho.
Isliye GCD le lo - largest number jo sabko divide kare.

$$\text{GCD}(16, 32, 96) = 16$$

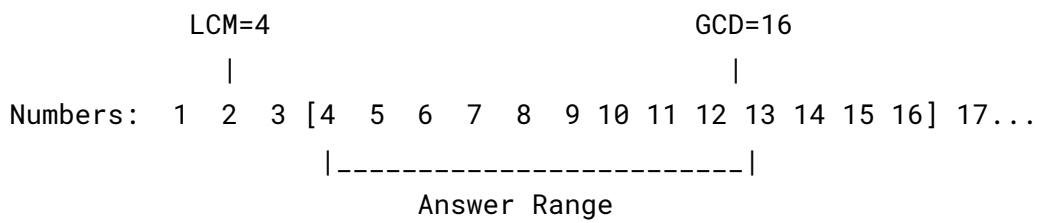
Iska matlab: Answer 16 se bada nahi ho sakta, aur 16 ka factor hona chahiye

Visualization - The Number Line

Array A $\rightarrow$ LCM $\rightarrow$ Starting Point (minimum value)

Array B → GCD → Ending Point (maximum value)

Example with $A=[2,4]$, $B=[16,32,96]$:



Candidates: Numbers between 4 and 16 that divide 16 perfectly
→ 4, 8, 16 (these are multiples of 4 AND factors of 16)

Key Insight: 💡

- LCM se start karo (ye minimum hai)
- GCD tak jao (ye maximum hai)
- Sirf wahi numbers valid hain jo GCD ko perfectly divide karein

Page 5: Dry Run - Step by Step

Example: $A = [2, 4]$, $B = [16, 32, 96]$

Step 1: Calculate LCM of Array A

```
# LCM nikalne ka formula: LCM(a,b) = (a*b) // GCD(a,b)
```

LCM of $[2, 4]$:

Step 1: $\text{LCM}(2, 4) = (2 * 4) // \text{GCD}(2, 4)$
 $= 8 // 2$
 $= 4$

Result: LCM_A = 4

Step 2: Calculate GCD of Array B

GCD nikalte hain pairwise

GCD of [16, 32, 96]:

Step 1: GCD(16, 32)

16 = 2×2×2×2

32 = 2×2×2×2×2

Common = 2×2×2×2 = 16

Step 2: GCD(16, 96)

16 = 2×2×2×2

96 = 2×2×2×2×2×3

Common = 2×2×2×2 = 16

Result: GCD_B = 16

Step 3: Find Numbers Between LCM and GCD

Range: 4 se lekar 16 tak

Condition: Number GCD_B ko perfectly divide kare (factor ho 16 ka)

Check each number:

4 → 16%4=0 ✓ (4 is multiple of 4, factor of 16)

5 → 16%5=1 ✗

6 → 16%6=4 ✗

7 → 16%7=2 ✗

8 → 16%8=0 ✓ (8 is multiple of 4, factor of 16)

9 → 16%9=7 ✗

10 → 16%10=6 ✗

11 → 16%11=5 ✗

12 → 16%12=4 ✗

13 → 16%13=3 ✗

14 → 16%14=2 ✗

15 → 16%15=1 ✗

16 → 16%16=0 ✓ (16 is multiple of 4, factor of 16)

Valid numbers: [4, 8, 16]

Count: 3 ✓

Step 4: Verify Each Valid Number

For answer = 4:

Array A: 4%2=0 ✓, 4%4=0 ✓

Array B: 16%4=0 ✓, 32%4=0 ✓, 96%4=0 ✓

For answer = 8:

Array A: 8%2=0 ✓, 8%4=0 ✓

Array B: 16%8=0 ✓, 32%8=0 ✓, 96%8=0 ✓

For answer = 16:

Array A: 16%2=0 ✓, 16%4=0 ✓

Array B: 16%16=0 ✓, 32%16=0 ✓, 96%16=0 ✓

All 3 numbers are valid! 🎉

Page 6: Python Implementation - Code Explained

Complete Optimized Solution

```
import math # GCD ke liye ready-made function

def getTotalX(a, b):
    # Step 1: LCM of all elements in array a
    lcm_a = a[0]
    for i in range(1, len(a)):
        # LCM(a,b) = (a * b) // GCD(a,b)
        lcm_a = (lcm_a * a[i]) // math.gcd(lcm_a, a[i])

    # Step 2: GCD of all elements in array b
    gcd_b = b[0]
    for i in range(1, len(b)):
        gcd_b = math.gcd(gcd_b, b[i])

    # Step 3: Count multiples of lcm_a that divide gcd_b
    count = 0
    current = lcm_a # LCM se start karo

    while current <= gcd_b: # GCD tak check karo
        if gcd_b % current == 0: # Kya current GCD ka factor hai?
            count += 1 # Ha! Ek valid number mil gaya
            current += lcm_a # Next multiple of LCM

    return count

# Test the function
a = [2, 4]
b = [16, 32, 96]
result = getTotalX(a, b)
print(f"Total valid numbers: {result}") # Output: 3
```

Line-by-Line Explanation 🧑🏫🏠

import math

- Python ka math module import karte hain
- Isme gcd() function milta hai (GCD nikalne ke liye)
- Khud ka GCD function likhne ki zaroorat nahi!

lcm_a = (lcm_a * a[i]) // math.gcd(lcm_a, a[i])

- LCM formula: $(\text{number1} \times \text{number2}) \div \text{GCD}(\text{number1}, \text{number2})$
- // integer division hai (decimal nahi chahiye)

- Example: $(4 \times 6) // 2 = 24 // 2 = 12$

```
gcd_b = math.gcd(gcd_b, b[i])
```

- GCD nikalo current aur next element ka
- Pairwise GCD nikalte jaao
- Finally sabka common GCD mil jaayega

```
current += lcm_a
```

- Hum LCM ke multiples generate kar rahe hain
- lcm_a, 2×lcm_a, 3×lcm_a, ...
- Ye sab Array A ke har element ke multiples honge guaranteed!

```
gcd_b % current == 0
```

- Check kar rahe hain ki current number GCD ko divide karta hai ya nahi
- Agar remainder 0 aaya, matlab valid number hai

Page 7: Code Evolution - Beginner to Pro

Beginner Approach (Brute Force)

```
def getTotalX_bruteforce(a, b):
    count = 0

    # Array A ka max se lekar Array B ka min tak check karo
    for num in range(max(a), min(b) + 1):
        # Check 1: Kya num, Array A ke har element ka multiple hai?
        condition1 = True
        for ele_a in a:
            if num % ele_a != 0: # num, ele_a se divisible nahi
                condition1 = False
                break

        if not condition1:
            continue # Agla number check karo

        # Check 2: Kya num, Array B ke har element ka factor hai?
        condition2 = True
        for ele_b in b:
            if ele_b % num != 0: # ele_b, num se divisible nahi
                condition2 = False
                break

        if condition2:
            count += 1 # Dono conditions pass!

    return count
```


Problems with Brute Force: ❌

- Range bohot badi ho sakti hai (1 se 1000000 tak!)
- Har number ke liye poori arrays check kar rahe hain
- Time Complexity: $O(\text{range} \times (\text{len}(a) + \text{len}(b))) \rightarrow$ Bohot slow!

🚀 Pro Approach (LCM + GCD)

```
def getTotalX_optimized(a, b):  
    # LCM and GCD calculate karo (O(n))  
    lcm_a = calculate_lcm(a)  
    gcd_b = calculate_gcd(b)  
  
    # Sirf LCM ke multiples check karo up to GCD  
    count = 0  
    for multiple in range(lcm_a, gcd_b + 1, lcm_a):  
        if gcd_b % multiple == 0:  
            count += 1  
  
    return count
```

Why Better? ✅

- Sirf LCM ke multiples check ho rahe hain (bohot kam numbers!)
- Array A ka check karne ki zaroorat nahi (LCM guarantee deta hai)
- Array B ka full check nahi, sirf GCD se divide check
- Time Complexity: $O(\log n + \text{gcd_b}/\text{lcm_a}) \rightarrow$ Lightning fast!

Performance Comparison:

A = [2, 4], B = [16, 32, 96]

Brute Force: 13 numbers check × 5 array elements = 65 operations
Optimized: 4 numbers check × 1 condition = 4 operations

Speed boost: 16x faster! 🏃🏻➡️

Page 8: Complexity Analysis

🕒 Time Complexity

Operation	Complexity	Explanation
LCM of array A	$O(n \times \log m)$	n elements, m is largest number
GCD of array B	$O(m \times \log k)$	m elements, k is largest number

Operation	Complexity	Explanation
Main loop	$O(\text{GCD}/\text{LCM})$	Number of multiples to check
Total	$O(n + m + k)$	Where $k = \text{GCD}/\text{LCM}$

Simple Words Mein:

- LCM nikalne mein time lagta hai array ke size ke hisaab se
- GCD nikalne mein bhi same
- Main loop mein GCD/LCM jitne steps (usually very small number)

Real Example:

```
A = [2, 4] → LCM = 4 (2 calculations)
B = [16, 32, 96] → GCD = 16 (2 calculations)
Loop runs: 4, 8, 12, 16 → 4 iterations
Total work: ~8 operations
```

 Space Complexity

Component	Space	Explanation
Variables	$O(1)$	Sirf lcm_a, gcd_b, count store kar rahe
Arrays	$O(1)$	Original arrays already given hain
Total	$O(1)$	Constant space, kitna bhi data ho!

Simple Words Mein:

- Koi extra memory nahi use kar rahe
- Sirf 3-4 variables
- Arrays bhi same reference use kar rahe hain

Page 9: Common Mistakes & Solutions

 Mistake 1: Factor aur Multiple Confuse Karna

```
# ❌ WRONG - Confusion
# "Kya 4, 12 ka multiple hai?"
# Answer: No! 4 is a FACTOR of 12, 12 is MULTIPLE of 4

# ✓ CORRECT
def check_factor(num, target):
    return target % num == 0 # num, target ka factor hai kya?
```

```
def check_multiple(num, base):  
    return num % base == 0 # num, base ka multiple hai kya?
```

Memory Trick: 🧠

- Factor: “F” se “Fraction” → Chhota part (divides bigger)
- Multiple: “M” se “Multiply” → Bada number (multiplication se bana)

🚫 Mistake 2: Wrong LCM Implementation

```
# ❌ WRONG - Simple multiplication  
lcm = a * b # This is often too big!  
  
# ❌ WRONG - Wrong formula  
lcm = math.gcd(a, b) // (a * b) # Ula formula!  
  
# ✓ CORRECT  
lcm = (a * b) // math.gcd(a, b) # Perfect!
```

🚫 Mistake 3: Forgetting Range End (+1)

```
gcd_b = 16  
  
# ❌ WRONG - Missing last number  
for num in range(lcm_a, gcd_b): # Stops at 15!  
    # 16 kabhi check nahi hoga!  
  
# ✓ CORRECT  
for num in range(lcm_a, gcd_b + 1): # Includes 16  
    # Perfect, 16 bhi check hoga
```

🚫 Mistake 4: Incorrect Modulus Direction

```
# ❌ WRONG - Modulus direction wrong  
if current % gcd_b == 0: # Ye check karega "kya current, gcd_b ka multiple hai?"  
    # But humein chahiye "kya current, gcd_b ka factor hai?"  
  
# ✓ CORRECT  
if gcd_b % current == 0: # Check: kya current GCD ko divide karta hai?  
    count += 1
```

🚫 Mistake 5: Starting from Wrong Point

```
# ❌ WRONG - Starting from 0
current = 0 # 0 se start? 0 se division possible nahi!

# ❌ WRONG - Starting from 1
current = 1 # 1 Array A ka LCM multiple nahi ho sakta!

# ✓ CORRECT
current = lcm_a # LCM se start karo, perfect!
```

Page 10: Practice Problems

Easy Level (Start Here)

1. Simple Factor Check

Problem: Array A = [2], Array B = [10]
Find: Kitne numbers hain jo 2 ke multiple hain aur 10 ke factor?
Hint: LCM(2) = 2, GCD(10) = 10, Check multiples of 2 up to 10

2. Two Elements Each

Problem: A = [3, 6], B = [12, 24]
Find: Count valid numbers
Hint: LCM of A = 6, GCD of B = 12, valid numbers: 6, 12

3. Same Numbers

Problem: A = [4], B = [4]
Find: Count
Hint: LCM = 4, GCD = 4, sirf 4 hi check karna

Medium Level (Practice More)

4. Three Elements

Problem: A = [2, 3, 6], B = [42, 84]
Find: Count
Hint: LCM(2,3,6) = 6, GCD(42,84) = 42, check multiples of 6

5. Larger Numbers

Problem: A = [2, 4, 8], B = [16, 32, 48]
Find: Count
Hint: LCM = 8, GCD = 16, valid: 8, 16

6. Single Element Array

Problem: A = [1], B = [100]
Find: Count
Hint: LCM = 1, GCD = 100, all factors of 100 work!

Interview Level (Challenge) 💪

7. Prime Numbers

Problem: A = [2, 3, 5, 7], B = [210, 420, 630]
Find: Count
Hint: LCM = 210, GCD = 210, only one number!

8. Multiple Valid Numbers

Problem: A = [2, 4], B = [16, 32, 64, 96]
Find: Count
Hint: GCD(16,32,64,96) = 16, LCM(2,4) = 4, check: 4,8,16

9. Edge Case - No Solution

Problem: A = [3, 5], B = [4, 8]
Find: Count
Hint: LCM(3,5)=15 > min(B)=4, no solution possible (return 0)

10. Large Array Test

Problem: A = [2, 4, 8, 16, 32], B = [64, 128, 256, 512]
Find: Count efficiently
Hint: LCM of A = 32, GCD of B = 64, multiples: 32, 64

Page 11: Final Revision Notes 📝

🎯 Important Formulas

Formula	Use Case	Example
$LCM(a,b) = (a \times b) // GCD(a,b)$	Calculate LCM	$LCM(4,6) = 24 // 2 = 12$
$GCD(a,b) = \text{math.gcd}(a,b)$	Calculate GCD	$GCD(12,18) = 6$
<code>num % divisor == 0</code>	Check divisibility	$12 \% 4 = 0 \rightarrow 4 \text{ divides } 12$

 Pattern Recognition Tips

Kab GCD Use Karna Hai?

- ✓ Finding common factors
- ✓ “Sabse bada number jo sabko divide kare”
- ✓ Array ke numbers ke factors dhundhne hain
- ✓ Maximum limit set karni hai

Kab LCM Use Karna Hai?

- ✓ Finding common multiples
- ✓ “Sabse chhota number jo sabka multiple ho”
- ✓ Minimum starting point set karna hai
- ✓ Array ke numbers ke multiples generate karne hain

 Problem-Solving Template

```
def solve_between_two_arrays(a, b):
    # Step 1: LCM of first array → Starting point
    lcm = find_lcm(a)

    # Step 2: GCD of second array → Ending point
    gcd = find_gcd(b)

    # Step 3: If LCM > GCD, no solution
    if lcm > gcd:
        return 0

    # Step 4: Count multiples of LCM that divide GCD
    count = 0
    for i in range(lcm, gcd + 1, lcm):
        if gcd % i == 0:
            count += 1

    return count
```

 Memory Tricks

1. **"LCM → Left side"** (Starting point)
2. **"GCD → Right side"** (Ending point)
3. **"Multiple of LCM, Factor of GCD"** (Valid number condition)
4. **"LCM se chhota nahi, GCD se bada nahi"** (Range yaad rakho)

 Success Checklist

Jaane se pehle, confirm karo ki tumhe ye sab aata hai:

- ☐ Factor aur Multiple ka difference clear hai
- ☐ GCD nikalna aata hai (pairwise method)
- ☐ LCM nikalna aata hai (formula use karke)
- ☐ Brute force vs Optimized approach samajh aati hai
- ☐ Code line-by-line explain kar sakte ho
- ☐ Edge cases handle kar sakte ho (LCM > GCD)
- ☐ Kam se kam 5 practice problems solve ki hain

Bonus Tip

Interview Mein Kya Bolein:

1. "Pehle main brute force approach batata hoon..."
2. "Phir optimize karunga GCD-LCM use karke..."
3. "Time complexity $O(n \times \log m)$ se $O(\log n)$ ho jaayegi"

Real-World Applications:

- Calendar/Event scheduling (LCM)
- Resource allocation (GCD)
- Network packet timing
- Gear ratio calculations

Congratulations!

Ab tum "Between Two Arrays" problem ke MASTER ho!

Yaad rakho:

"LCM se shuru karo, GCD tak jaao, multiples count karte jaao!"

Happy Coding! 💻 ✨